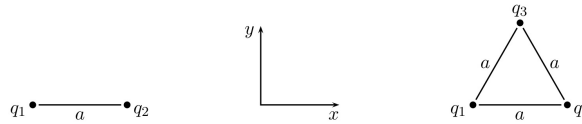


Opgave 1

Der optræder her tre punktladninger, $q_1 = -3,00 \mu\text{C}$, $q_2 = 2,00 \mu\text{C}$ og $q_3 = 4,00 \mu\text{C}$, hvor $1 \mu\text{C} = 10^{-6} \text{C}$.

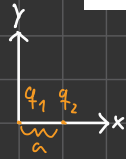


De følgende spørgsmål bedes besvaret først ved analytiske udtryk og dernæst ved indsættelse af talværdier. De efterspurte kræfter bedes givet på vektorform i forhold til det indtegnede xy -system.

- a) Først anbringes q_1 og q_2 begge langs x -aksen med en indbyrdes afstand på $a = 20,0 \text{ cm}$, som vist på figuren til venstre. Den tredje ladning optræder endnu ikke. Bestem kraften på q_1 .

- Analytisk
- Numerisk

$$\vec{r} = r \cdot \hat{r}$$

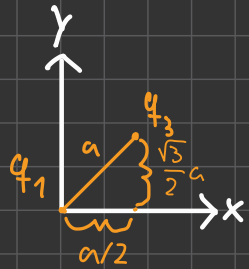


$$\vec{r} = (0 - a)\hat{x} = -a\hat{x}, \quad r = a, \quad \hat{r} = -\hat{x}$$

$$\vec{F}_a = \vec{F}_{q_2 \rightarrow q_1}$$

$$\vec{F}_a = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{a^2} (-\hat{x}) = \frac{1}{4\pi \cdot 8,85 \cdot 10^{-12} \text{C}^2/\text{N}\cdot\text{m}^2} \frac{-3 \cdot 10^{-6} \text{C} \cdot 2 \cdot 10^{-6} \text{C}}{(20 \cdot 10^{-2} \text{m})^2} (-\hat{x}) = \frac{6 \text{ N}\cdot\text{m}^2}{4,45 \text{ m}^2} \hat{x} = 1,35 \text{ N} \hat{x}$$

- b) Dernæst tilføjes q_3 , som vist på figuren til højre. De tre ladninger danner en ligesidet trekant. Bestem igen kraften på q_1 .



$$\vec{r} = (0 - \frac{a}{2})\hat{x} + (0 - \frac{\sqrt{3}}{2}a)\hat{y} = -\frac{a}{2}\hat{x} - \frac{\sqrt{3}}{2}a\hat{y}, \quad r = a, \quad \hat{r} = -\frac{\hat{x}}{2} - \frac{\sqrt{3}}{2}\hat{y}$$

$$\vec{F}_{q_3 \rightarrow q_1} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{a^2} \left(-\frac{\hat{x}}{2} - \frac{\sqrt{3}}{2}\hat{y} \right) = \frac{-3 \cdot 4 \text{ C}^2}{4,45 \text{ m}^2 \text{ C}^2/\text{N}\cdot\text{m}^2} \left(-\frac{\hat{x}}{2} - \frac{\sqrt{3}}{2}\hat{y} \right) = 2,70 \text{ N} \left(\frac{\hat{x}}{2} + \frac{\sqrt{3}}{2}\hat{y} \right) =$$

$$1,35 \text{ N} \hat{x} + 2,34 \text{ N} \hat{y}, \quad \vec{F}_b = \vec{F}_{q_3 \rightarrow q_1} + \vec{F}_{q_2 \rightarrow q_1} = 2,70 \text{ N} \hat{x} + 2,34 \text{ N} \hat{y}$$

Problem 2.1

- (a) Twelve equal charges, q , are situated at the corners of a regular 12-sided polygon (for instance, one on each numeral of a clock face). What is the net force on a test charge Q at the center?

Svarer til 12 ladninger ligeligt fordelt på en cirkel med radius r .

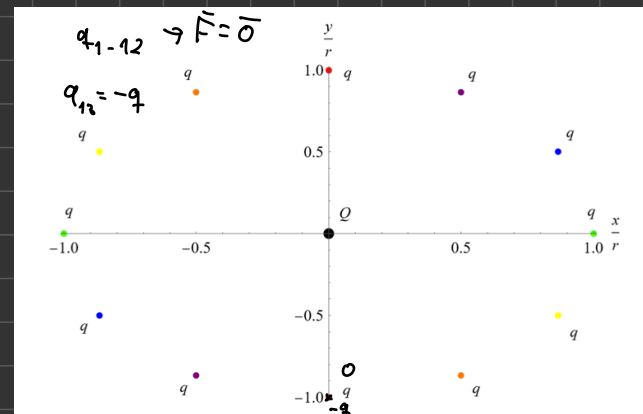
Positionvektoren for ladningen q_k er: $\vec{r}'_k = r \left(\cos \frac{2\pi k}{12} \hat{x} + \sin \frac{2\pi k}{12} \hat{y} \right)$, $0 \leq k \leq 11$

Alle kræfter er lige store og modsatrettede, så $\vec{F} = \vec{0}$.

$$\text{kan udregnes explicit } \vec{F} = \sum_{k=0}^{11} \frac{1}{4\pi\epsilon_0} \frac{q_k Q}{r_k^2} \hat{r}_k =$$

$$\frac{qQ}{4\pi\epsilon_0 r^2} \sum_{k=0}^{11} \left(\frac{\vec{0} - \vec{r}'_k}{r} \right) =$$

$$\frac{qQ}{4\pi\epsilon_0 r^3} r \left(\sum_{k=0}^{11} \cos \frac{2\pi k}{12} \hat{x} + \sum_{k=0}^{11} \sin \frac{2\pi k}{12} \hat{y} \right) = \dots = \vec{0}$$



(b) Suppose one of the 12 q 's is removed (the one at "6 o'clock"). What is the force on Q ? Explain your reasoning carefully.

At fjerne ladningen svarer til at introducere en ladning med størrelsen $-q$ oven i den gamle ladning. Kraften fra de 12 oprindelige ladninger er 0, så $F = \frac{qQ}{4\pi\epsilon_0 r^2}$ med retning mod den manglende ladning; i dette tilfælde $-\hat{y}$.

(c) Now 13 equal charges, q , are placed at the corners of a regular 13-sided polygon. What is the force on a test charge Q at the center?

Minder om a), men nu er $\vec{r}'_k = r \left(\cos \frac{2\pi k}{13} \hat{x} + \sin \frac{2\pi k}{13} \hat{y} \right)$, $0 \leq k \leq 12$.

Af symmetriårsager må $\vec{F} = 0$. Roterer vi ladningsfordelingen med vinklen $\Delta\theta = \frac{2\pi}{13}$ får vi en ladningsfordeling, der er identisk med den forrige.

Den eneste vektor, der forbliver uændret under en rotation $\Delta\theta \neq 0$ er $\vec{0}$.

(d) If one of the 13 q 's is removed, what is the force on Q ? Explain your reasoning.

Magen til b); $F = \frac{qQ}{4\pi\epsilon_0 r^2}$ med retning mod den manglende ladning.

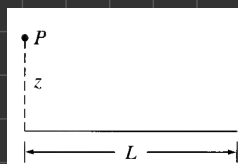


Figure 2.7

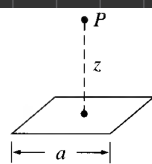


Figure 2.8

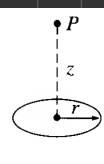


Figure 2.9

Problem 2.3 Find the electric field a distance z above one end of a straight line segment of length L (Fig. 2.7), which carries a uniform line charge λ . Check that your formula is consistent with what you would expect for the case $z \gg L$.

$$\vec{r} = z\hat{z}, \quad \vec{r}' = x\hat{x}, \quad d\vec{l}' = dx\hat{x}, \quad \vec{r} = \vec{r} - \vec{r}' = z\hat{z} - x\hat{x}, \quad r = \sqrt{z^2 + x^2}, \quad \hat{r} = \frac{z\hat{z} - x\hat{x}}{\sqrt{z^2 + x^2}}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int_0^L \frac{1}{r^2} \hat{r} dx = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{1}{z^2 + x^2} \frac{z\hat{z} - x\hat{x}}{\sqrt{z^2 + x^2}} dx = \frac{\lambda}{4\pi\epsilon_0} \left(\int_0^L \frac{z dx}{(z^2 + x^2)^{3/2}} \hat{z} - \int_0^L \frac{x dx}{(z^2 + x^2)^{3/2}} \hat{x} \right) = \frac{\lambda}{4\pi\epsilon_0} \left(\left[\frac{zx}{z^2\sqrt{x^2 + z^2}} \right]_0^L \hat{z} - \left[\frac{-1}{\sqrt{x^2 + z^2}} \right]_0^L \hat{x} \right) = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{L}{z\sqrt{L^2 + z^2}} \hat{z} + \left(\frac{1}{\sqrt{L^2 + z^2}} - \frac{1}{z} \right) \hat{x} \right)$$

$$\text{For } z \gg L \rightarrow \vec{E} = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{L}{z\sqrt{z^2}} \hat{z} + \left(\frac{1}{\sqrt{z^2}} - \frac{1}{z} \right) \hat{x} \right) = \frac{1}{4\pi\epsilon_0} \frac{\lambda L}{z^2} \hat{z}. \text{ Svarer til punktladning}$$

med $q = \lambda L$. Stemmer med intuition.

Problem 2.4 Find the electric field a distance z above the center of a square loop (side a) carrying uniform line charge λ (Fig. 2.8). [Hint: Use the result of Ex. 2.1.]

For hver af siderne svarer det til Ex. 2.2 med $L \rightarrow \frac{a}{2}$ og $z \rightarrow \sqrt{z^2 + (\frac{a}{2})^2}$ (afstanden fra midten af en side til p). For en side

$$\text{har vi da } E_1 = \frac{1}{4\pi\epsilon_0} \frac{2\lambda \frac{a}{2}}{\sqrt{z^2 + (\frac{a}{2})^2} \sqrt{z^2 + (\frac{a}{2})^2 + (\frac{a}{2})^2}} = \frac{1}{4\pi\epsilon_0} \frac{\lambda a}{\sqrt{z^2 + \frac{a^2}{4}} \sqrt{z^2 + \frac{a^2}{2}}}$$

Der 4 sider, og vi er kun interesseret i $\hat{z}$ -komponenten, da de andre udligner hinanden og dermed er 0. $\vec{E} = 4E_1 \cos \theta \hat{z}$, $\cos \theta = \frac{z}{\sqrt{z^2 + (\frac{a}{2})^2}}$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{4\lambda a}{\sqrt{z^2 + \frac{a^2}{4}} \sqrt{z^2 + \frac{a^2}{2}}} \frac{z}{\sqrt{z^2 + \frac{a^2}{4}}} \hat{z} = \frac{1}{4\pi\epsilon_0} \frac{4\lambda a z}{(z^2 + \frac{a^2}{4}) \sqrt{z^2 + \frac{a^2}{2}}} \hat{z}$$

Problem 1.15 Calculate the divergence of the following vector functions:

(a) $\mathbf{v}_a = x^2 \hat{x} + 3xz^2 \hat{y} - 2xz \hat{z}$.

(b) $\mathbf{v}_b = xy \hat{x} + 2yz \hat{y} + 3zx \hat{z}$.

(c) $\mathbf{v}_c = y^2 \hat{x} + (2xy + z^2) \hat{y} + 2yz \hat{z}$.

a) $\vec{\nabla} \cdot \vec{v}_a = \frac{\partial}{\partial x} x^2 + \frac{\partial}{\partial y} 3xz^2 - \frac{\partial}{\partial z} 2xz = 2x - 2x = 0$

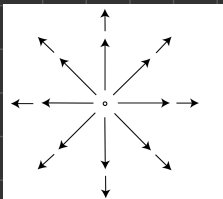
b) $\vec{\nabla} \cdot \vec{v}_b = \frac{\partial}{\partial x} xy + \frac{\partial}{\partial y} 2yz + \frac{\partial}{\partial z} 3zx = y + 2z + 3x$

c) $\vec{\nabla} \cdot \vec{v}_c = \frac{\partial}{\partial x} y^2 + \frac{\partial}{\partial y} (2xy + z^2) + \frac{\partial}{\partial z} 2yz = 2x + 2y$

Problem 1.16 Sketch the vector function

$$\mathbf{v} = \frac{\hat{\mathbf{r}}}{r^2},$$

and compute its divergence. The answer may surprise you... can you explain it?



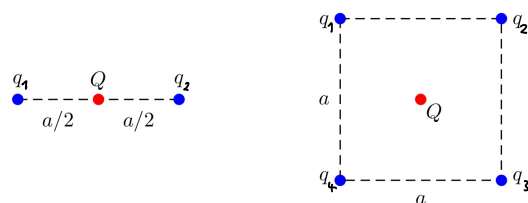
Bruger sfæriske koordinater, $\vec{\nabla} \cdot \vec{v} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{1}{r^2} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} 1 = 0$.

$\vec{\nabla} \cdot \vec{v} = 0$ gælder kun for $r \neq 0$ og det forklarer modstriden mellem

tegningen - hvor feltet tydeligvis divergerer væk fra origo - og resultat $\vec{\nabla} \cdot \vec{v} = 0$.

Opgave 1

Vi betragter i første omgang situationen på figuren til venstre, hvor to identiske ladninger q ligger i afstanden a fra hinanden, og ladningen Q ligger midt på forbindelseslinjen mellem de to.



- a) Bestem Q udtrykt ved q , således at systemet er i ligevægt, og kraften på hver af ladningerne er nul. Argumentér for, hvorvidt der er tale om en stabil eller en ustabil ligevægtstilstand.

$$\vec{F}_{q_1} = \frac{1}{4\pi\epsilon_0} \left(\frac{qQ}{(a/2)^2} (-\hat{x}) - \frac{q^2}{a^2} \hat{x} \right). \vec{F}_{q_1} = \vec{0} \Rightarrow \frac{qQ}{(a/2)^2} (-\hat{x}) - \frac{q^2}{a^2} \hat{x} = 0 \Rightarrow \frac{qQ}{a^2/4} = \frac{q^2}{a^2} \Rightarrow 4Q = -q \Rightarrow Q = -\frac{q}{4}$$

$$\vec{F}_Q = \vec{0}. \text{ Symmetri}$$

$$\vec{F}_{q_2} = \frac{1}{4\pi\epsilon_0} \left(\frac{qQ}{(a/2)^2} \hat{x} + \frac{q^2}{a^2} \hat{x} \right). Q = -\frac{q}{4} \Rightarrow \vec{F}_{q_2} = \frac{1}{4\pi\epsilon_0} \left(\frac{-4q^2}{4a^2} \hat{x} + \frac{q^2}{a^2} \hat{x} \right) = \vec{0}$$

Ustabil ligevægtstilstand s. 118 Earnshaws teorem: A charged particle cannot be held in a stable equilibrium by electrostatic forces alone

Vi betragter derefter situationen på figuren til højre, hvor fire identiske ladninger q danner et kvadrat med sidelængden a , og ladningen Q ligger i kvadratets midtpunkt.

- b) Bestem igen Q udtrykt ved q , således at systemet er i ligevægt, og kraften på hver af ladningerne er nul.

$$\text{Fra } q_3 \text{ til } q_1, \vec{r} = a\hat{y} - a\hat{x}, r = \sqrt{a^2 + a^2} = \sqrt{2}a, \hat{r} = \frac{-\hat{x}a + \hat{y}a}{\sqrt{2}a} = \frac{-\hat{x} + \hat{y}}{\sqrt{2}}$$

$$\text{Fra } Q \text{ til } q_1, \vec{r} = a\hat{y} - \frac{a\hat{x}}{2} - \frac{a\hat{y}}{2} = \frac{a}{2}(\hat{y} - \hat{x}), r = \sqrt{\frac{a^2}{4} + \frac{a^2}{4}} = \frac{a}{\sqrt{2}}, \hat{r} = \frac{a}{2} \frac{(-\hat{x} + \hat{y})}{a/\sqrt{2}} = \frac{-\hat{x} + \hat{y}}{\sqrt{2}} \checkmark$$

$$\vec{F}_{q_2 q_1} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a^2} (-\hat{x}). \vec{F}_{q_3 q_1} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a^2} \hat{y}. \vec{F}_{q_4 q_1} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{2a^2} \left(\frac{-\hat{x} + \hat{y}}{\sqrt{2}} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{2^{3/2}a^2} (-\hat{x} + \hat{y})$$

$$\vec{F}_{Q q_1} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{a^2/2} \left(\frac{-\hat{x} + \hat{y}}{\sqrt{2}} \right) = \frac{1}{4\pi\epsilon_0} \frac{\sqrt{2}qQ}{a^2} (-\hat{x} + \hat{y}). \vec{F}_{q_1} = \frac{1}{4\pi\epsilon_0} \left(\frac{q^2}{a^2} (-\hat{x}) + \frac{q^2}{a^2} \hat{y} + \frac{q^2}{2^{3/2}a^2} (-\hat{x} + \hat{y}) + \frac{\sqrt{2}qQ}{a^2} (-\hat{x} + \hat{y}) \right) = \frac{1}{4\pi\epsilon_0} \left[\left(\frac{q^2}{a^2} + \frac{q^2}{2^{3/2}a^2} + \frac{\sqrt{2}qQ}{a^2} \right) (-\hat{x}) + \left(\frac{q^2}{a^2} + \frac{q^2}{2^{3/2}a^2} + \frac{\sqrt{2}qQ}{a^2} \right) \hat{y} \right]. \vec{F}_{q_1} = \vec{0} \Rightarrow q^2 + \frac{q^2}{2^{3/2}} + \sqrt{2}qQ = 0 \Rightarrow$$

$$\sqrt{2}Q = -q - \frac{q}{2^{3/2}} = -q \left(1 + \frac{1}{2^{3/2}} \right) \Rightarrow Q = -q \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2} \cdot 2^{3/2}} \right) = -q \left(\frac{2\sqrt{2}}{4} + \frac{1}{4} \right) = -\frac{2\sqrt{2} + 1}{4} q$$

Problem 2.5 Find the electric field a distance z above the center of a circular loop of radius r (Fig. 2.9), which carries a uniform line charge λ .

Feltet: xy -planen er nul grundet symmetri - det udligner sig selv.

$$r = \sqrt{r^2 + z^2}, \cos \theta = \frac{z}{r}. \vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} dl \cos \theta \hat{z}. \int dl = 2\pi r \rightarrow \text{omkreds af en cirkel.}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{12\pi r z}{(r^2 + z^2)^{3/2}} \hat{z}$$

Problem 2.6 Find the electric field a distance z above the center of a flat circular disk of radius R (Fig. 2.10), which carries a uniform surface charge σ . What does your formula give in the limit $R \rightarrow \infty$? Also check the case $z \gg R$.

En disk svarer til ringe af radius r og tykkelse dr . Vi kender feltet af en enkelt ring fra opg. 2.5. Den totale ladning af én ring er $\sigma 2\pi r dr$. Så "line charge" pr. ring må være $\lambda = \sigma dr$.

$$\vec{E}_{\text{ring}} = \frac{1}{4\pi\epsilon_0} \frac{\sigma dr 2\pi r z}{(r^2 + z^2)^{3/2}} \hat{z}. \quad \vec{E}_{\text{disk}} = \frac{1}{4\pi\epsilon_0} \sigma 2\pi z \int_0^R \frac{r}{(r^2 + z^2)^{3/2}} dr \hat{z} = \frac{1}{4\pi\epsilon_0} \sigma 2\pi z \left[\frac{1}{z} - \frac{1}{\sqrt{z^2 + R^2}} \right] \hat{z}$$

Alternativt: $\vec{r} = z \hat{z}$, $\vec{r}' = r(\cos\phi \hat{x} + \sin\phi \hat{y})$, $\vec{r} = \vec{r} - \vec{r}' = -r(\cos\phi \hat{x} + \sin\phi \hat{y}) + z \hat{z}$

$$r = \sqrt{r^2 \cos^2\phi + r^2 \sin^2\phi + z^2} = \sqrt{r^2 + z^2}, \quad \hat{r} = \frac{-r(\cos\phi \hat{x} + \sin\phi \hat{y}) + z \hat{z}}{\sqrt{r^2 + z^2}} \quad da = r dr d\phi$$

$$(2.7) \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^R \frac{\sigma}{r^2 + z^2} \frac{-r(\cos\phi \hat{x} + \sin\phi \hat{y}) + z \hat{z}}{\sqrt{r^2 + z^2}} r dr d\phi$$

Ved beregning fås $E_x = E_y = 0$, dette ses også ved symmetri.

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^R \frac{\sigma z r}{(r^2 + z^2)^{3/2}} dr d\phi \hat{z} = \frac{2\pi\sigma z}{4\pi\epsilon_0} \int_0^R \frac{r dr}{(r^2 + z^2)^{3/2}} \hat{z} = \frac{1}{4\pi\epsilon_0} \sigma 2\pi z \left[\frac{1}{z} - \frac{1}{\sqrt{z^2 + R^2}} \right] \hat{z}$$

$$R \rightarrow \infty \Rightarrow \frac{1}{4\pi\epsilon_0} \sigma 2\pi z \left[\frac{1}{z} - 0 \right] \hat{z} = \frac{1}{4\pi\epsilon_0} \sigma 2\pi \hat{z} = \frac{\sigma}{2\epsilon_0} \hat{z} \rightarrow \text{felt fra en uendelig flade (2.17)}$$

$$\frac{1}{\sqrt{1+x}} \approx 1 - \frac{1}{2}x$$

$$z \gg R \Rightarrow \frac{1}{\sqrt{z^2 + R^2}} = \frac{1}{z \sqrt{1 + R^2/z^2}} = \frac{1}{z} \frac{1}{\sqrt{1 + R^2/z^2}} \approx \frac{1}{z} \left(1 - \frac{R^2}{2z^2} \right) = \frac{1}{z} - \frac{R^2}{2z^3}$$

$$\vec{E}_{\text{disk}} \Rightarrow \frac{1}{4\pi\epsilon_0} \sigma 2\pi z \left[\frac{1}{z} - \left(\frac{1}{z} - \frac{R^2}{2z^3} \right) \right] \hat{z} = \frac{1}{4\pi\epsilon_0} \frac{\sigma 2\pi z R^2}{2 z^3} = \frac{1}{4\pi\epsilon_0} \frac{\sigma \pi R^2}{z^2} \quad \text{Svarer til feltet}$$

fra en punktladning med $Q = \sigma \pi R^2$.

Problem 2.9 Suppose the electric field in some region is found to be $\mathbf{E} = kr^3\hat{\mathbf{r}}$, in spherical coordinates (k is some constant).

(a) Find the charge density ρ .

(2.14)

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 kr^3) = \frac{k}{r^2} \frac{\partial}{\partial r}(r^5) = \frac{5k}{r^2} r^4 = 5kr^2 = \frac{\rho}{\epsilon_0} \Rightarrow \underline{\rho = 5k\epsilon_0 r^2}$$

(b) Find the total charge contained in a sphere of radius R , centered at the origin. (Do it two different ways.)

(2.13)

$$\oint \vec{E} \cdot d\vec{a} = kR^3 4\pi R^2 = 4k\pi R^5 = \frac{Q_{enc}}{\epsilon_0} \Rightarrow Q_{enc} = \underline{4k\pi\epsilon_0 R^5}$$

$$Q_{enc} = \int_V \rho d\tau = \int_0^{2\pi} \int_0^\pi \int_0^R 5k\epsilon_0 r^2 r^2 \sin\theta dr d\theta d\phi = 5k\epsilon_0 \frac{R^5}{5} 2 \cdot 2\pi = \underline{4k\pi\epsilon_0 R^5}$$

Problem 1.11 Find the gradients of the following functions:

(a) $f(x, y, z) = x^2 + y^3 + z^4$.

(b) $f(x, y, z) = x^2 y^3 z^4$.

(c) $f(x, y, z) = e^x \sin(y) \ln(z)$.

a) $\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z} = \underline{2x\hat{x} + 3y^2\hat{y} + 4z^3\hat{z}}$

b) $\vec{\nabla} f = \underline{2xy^3z^4\hat{x} + 3x^2y^2z^4\hat{y} + 4x^2y^3z^3\hat{z}}$

c) $\vec{\nabla} f = \underline{e^x \sin y \ln z \hat{x} + e^x \cos y \ln z \hat{y} + e^x \sin y \frac{1}{z} \hat{z}}$

Problem 1.18 Calculate the curls of the vector functions in Prob. 1.15.

(a) $\mathbf{v}_a = x^2 \hat{\mathbf{x}} + 3xz^2 \hat{\mathbf{y}} - 2xz \hat{\mathbf{z}}$.

(b) $\mathbf{v}_b = xy \hat{\mathbf{x}} + 2yz \hat{\mathbf{y}} + 3zx \hat{\mathbf{z}}$.

(c) $\mathbf{v}_c = y^2 \hat{\mathbf{x}} + (2xy + z^2) \hat{\mathbf{y}} + 2yz \hat{\mathbf{z}}$.

a) $\vec{\nabla} \times \vec{v}_a = \left(\frac{\partial}{\partial y}(-2xz) - \frac{\partial}{\partial z}(3xz^2) \right) \hat{x} + \left(\frac{\partial}{\partial z}x^2 - \frac{\partial}{\partial x}(-2xz) \right) \hat{y} + \left(\frac{\partial}{\partial x}(3xz^2) - \frac{\partial}{\partial y}x^2 \right) \hat{z} = \underline{-6xz\hat{x} + 2z\hat{y} + 3z\hat{z}}$

b) $\vec{\nabla} \times \vec{v}_b = \underline{-2y\hat{x} - 3z\hat{y} - x\hat{z}}$

c) $\vec{\nabla} \times \vec{v}_c = (2z - 2z)\hat{x} + (2y - 2y)\hat{y} + (2y - 2y)\hat{z} = \underline{\vec{0}}$

Problem 1.33 Test Stokes' theorem for the function $\mathbf{v} = (xy)\hat{\mathbf{x}} + (2yz)\hat{\mathbf{y}} + (3zx)\hat{\mathbf{z}}$, using the triangular shaded area of Fig. 1.34.

$\int_0^2 \int_0^{2-z}$ giver firksam

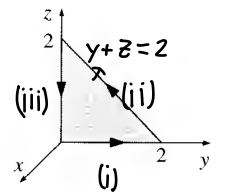


Figure 1.34

$$\bar{\nabla} \times \bar{v} = -2y\hat{x} - 3z\hat{y} - x\hat{z}. \quad d\bar{a} = dydz\hat{x}. \quad x=0 \quad (\bar{\nabla} \times \bar{v}) \cdot d\bar{a} = -2y dy dz.$$

$$\int (\bar{\nabla} \times \bar{v}) \cdot d\bar{a} = \int_0^2 \int_0^{2-z} (-2y) dy dz = \int_0^2 -(2-z)^2 dz = \int_0^2 -4 - z^2 + 4z dz =$$

$$\left[-4z - \frac{z^3}{3} + 2z^2 \right]_0^2 = -8 - \frac{8}{3} + 8 = -\frac{8}{3}. \quad \text{Alternativt: } \int (\bar{\nabla} \times \bar{v}) \cdot d\bar{a} = \int_0^2 \int_0^{2-y} (-2y) dz dy =$$

$$\int_0^2 -2y(2-y) dy = \int_0^2 -4y + 2y^2 dy = \left[-2y^2 + \frac{2}{3}y^3 \right]_0^2 = -8 + \frac{16}{3} = -\frac{8}{3}$$

Linje integrale : 3 dele: (i) $x=z=0, d\bar{l} = dy\hat{y}. \bar{v} \cdot d\bar{l} = 0.$

(ii) $x=0, z=2-y, dz = -dy, y: 2 \rightarrow 0. d\bar{l} = dy\hat{y} - dy\hat{z}. \bar{v} \cdot d\bar{l} = 2y(2-y)dy = (4y - 2y^2)dy$

$$\int \bar{v} \cdot d\bar{l} = \int_2^0 (4y - 2y^2) dy = \left[2y^2 - \frac{2}{3}y^3 \right]_2^0 = -8 + \frac{16}{3} = -\frac{8}{3}. \quad \text{(iii) } x=y=0, d\bar{l} = dz\hat{z}. \bar{v} \cdot d\bar{l} = 0.$$

$$\oint \bar{v} \cdot d\bar{l} = 0 - \frac{8}{3} + 0 = -\frac{8}{3} \Rightarrow \int (\bar{\nabla} \times \bar{v}) \cdot d\bar{a} = \oint \bar{v} \cdot d\bar{l}$$

Problem 1.39 Compute the divergence of the function

$$\mathbf{v} = (r \cos \theta) \hat{\mathbf{r}} + (r \sin \theta) \hat{\boldsymbol{\theta}} + (r \sin \theta \cos \phi) \hat{\boldsymbol{\phi}}.$$

Check the divergence theorem for this function, using as your volume the inverted hemispherical bowl of radius R , resting on the xy plane and centered at the origin (Fig. 1.40).

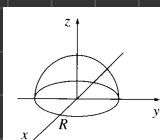


Figure 1.40

$$\bar{\nabla} \cdot \bar{\mathbf{v}} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 r \cos \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta r \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (r \sin \theta \cos \phi) = 3 \cos \theta + \frac{2 \sin \theta \cos \theta}{\sin \theta} +$$

$$- \sin \phi = 5 \cos \theta - \sin \phi. \quad \text{Div. theorem} \quad \int (\bar{\nabla} \cdot \bar{\mathbf{v}}) d\tau = \oint \bar{\mathbf{v}} \cdot d\bar{\mathbf{a}}$$

$$\int (\bar{\nabla} \cdot \bar{\mathbf{v}}) d\tau = \int_0^{2\pi} \int_0^{\pi/2} \int_0^R (5 \cos \theta - \sin \phi) r^2 \sin \theta dr d\theta d\phi = \frac{R^3}{3} \int_0^{2\pi} \int_0^{\pi/2} (5 \cos \theta - \sin \phi) \sin \theta d\theta d\phi =$$

$$\frac{R^3}{3} \int_0^{2\pi} \left(\frac{-2 \sin \phi - 5}{2} \right) d\phi = \frac{R^3}{3} 5\pi = \frac{5\pi}{3} R^3$$

To overflader - halvkuglen: $d\bar{\mathbf{a}} = R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}; r = R; \theta: 0 \rightarrow \frac{\pi}{2}; \phi: 0 \rightarrow 2\pi$

$$\oint \bar{\mathbf{v}} \cdot d\bar{\mathbf{a}} = \int_0^{2\pi} \int_0^{\pi/2} R \cos \theta R^2 \sin \theta d\theta d\phi = R^3 \frac{1}{2} 2\pi = \pi R^3 \quad \hat{\boldsymbol{\theta}} = -\hat{\mathbf{z}}, \text{ när } \theta = \frac{\pi}{2}$$

Bunden: $d\bar{\mathbf{a}} = dr r \sin \theta d\phi \hat{\boldsymbol{\theta}} \stackrel{\theta = \frac{\pi}{2}}{\downarrow} = r dr d\phi \hat{\boldsymbol{\theta}}; r: 0 \rightarrow R; \phi: 0 \rightarrow 2\pi.$

$$\oint \bar{\mathbf{v}} \cdot d\bar{\mathbf{a}} = \int_0^{2\pi} \int_0^R r \sin \theta r dr d\phi = 2\pi \frac{R^3}{3} \cdot 2\pi \frac{R^3}{3} + \pi R^3 = \frac{5\pi}{3} R^3 = \int (\bar{\nabla} \cdot \bar{\mathbf{v}}) d\tau \quad \checkmark$$